

Recitation 02
Coordinate Transform and Lyapunov Stability

Exercise 1

Consider the following linear discrete-time system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k), \\y(k) &= Cx(k) + Du(k),\end{aligned}$$

and assume A allows an eigendecomposition¹ $A = W\Delta W^{-1}$, where $W = [w_1, \dots, w_n]$ is the matrix of eigenvectors w_i , and $\Delta = \text{diag}([\lambda_1, \dots, \lambda_n])$ is the corresponding vector of eigenvalues λ_i .

1. Find a coordinate transformation $\tilde{x}(k) = Tx(k)$ such that $\tilde{x}(k+1) = \tilde{A}\tilde{x}(k) + \tilde{B}u(k)$ with diagonal $\tilde{A}$. Find the expressions for $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ and $\tilde{D}$.

Consider now the autonomous system, i.e., $u(k) = 0$ for all $k \geq 0$, and assume that the absolute value of all eigenvalues of A are $|\lambda_i| < 1$.

2. Show that the infinite horizon cost $\sum_{k=0}^{\infty} x(k)^T Q x(k)$ is bounded for all $Q > 0$ (Q positive definite).

Hints:

- A symmetric real matrix Q is positive definite if $z^T Q z > 0$ for every column vector $z \neq 0$
- Every matrix $Q > 0$ can be decomposed as $Q = Q^{\frac{1}{2}} Q^{\frac{1}{2}}$ with $Q^{\frac{1}{2}} > 0$
- $\|Ax\|_2 \leq \|A\|_2 \|x\|_2 = \sqrt{\lambda_{\max}(A^T A)} \|x\|_2$, where $\lambda_{\max}(A^T A)$ is the maximum eigenvalue of $A^T A$
- $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$ for $|r| < 1$

3. Show that for every $Q > 0$ there exists $P > 0$ satisfying the Lyapunov equation

$$\underbrace{x(k+1)^T P x(k+1)}_{V(x(k+1))} - \underbrace{x(k)^T P x(k)}_{V(x(k))} = - \underbrace{x(k)^T Q x(k)}_{\alpha(x(k))}.$$

Exercise 2

This exercise investigates stability properties of the origin for the following nonlinear discrete-time system

$$x(k+1) = \underbrace{\frac{x(k) + \sin(x(k))}{4}}_{:=f(x)}. \quad (1)$$

1. What can be concluded about stability of the equilibrium point $x_s = 0$ for system (1) using Lyapunov's **indirect** method?
2. What can be concluded about stability of the equilibrium point $x_s = 0$ for system (1) using Lyapunov's **direct** method?

¹i.e. $A \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors. The statements in 2) and 3) hold regardless of this assumption.